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Regulating Consumer Financial Products: Evidence from Credit Cards

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1 Big picture

- **Question.** Can consumer financial regulation reduce borrowing costs, or do firms simply undo the regulation by raising other prices and restricting credit? The paper studies this question using the 2009 Credit Card Accountability Responsibility and Disclosure Act, usually called the **CARD Act**.
- **Setting.** The market is U.S. credit cards from 2008 to 2011. The policy changed how issuers could charge credit card fees and how they had to disclose repayment information on monthly statements.
- **Core policy treatments.** The paper studies two main pieces of the CARD Act:
 1. limits on fees, especially over-limit fees and late fees;
 2. a disclosure nudge that tells borrowers the payment needed to pay off the balance in 36 months.
- **Main empirical problem.** The CARD Act was implemented during and after the financial crisis. A simple before-after comparison would mix regulation with macroeconomic shocks, changes in bank risk, consumer deleveraging, and credit-cycle effects.
- **Identification strategy.** The authors use a difference-in-differences design. Consumer credit cards were covered by the CARD Act and form the treatment group. Small business credit cards were not covered and form the control group. The identifying assumption is that, absent the law, outcomes for consumer and small business cards would have followed parallel trends.
- **Why small business cards are a useful control.** Small business cards are institutionally close to consumer cards. Both are typically backed by the personal assets of the cardholder, both are screened using personal FICO scores, and both are often managed by the same bank credit card divisions. The legal difference is that small business cards were excluded from the CARD Act.
- **Main result on fees.** The fee restrictions sharply reduced fees. For borrowers with FICO scores below 660, total fees fell by about 5.3 percentage points of average daily balances. For borrowers with FICO scores above 660, total fees fell by about 0.5 percentage points of average daily balances.
- **Aggregate magnitude.** Combining low- and high-FICO accounts, the authors estimate that the fee rules reduced borrowing costs by an annualized 1.6% of average daily balances. Scaled to the national credit card market, this implies consumer savings of about \$11.9 billion per year.
- **Main result on offsets.** The paper finds no evidence that issuers offset the fee reduction by raising interest charges, increasing other fees, increasing interchange income, or cutting credit supply. The preferred estimate of the interest-charge offset is approximately zero, and the authors can rule out an offset greater than 61% at the 95% confidence level.
- **Main result on the nudge.** The repayment disclosure had a small behavioral effect. The share of accounts making payments close to the 36-month payoff amount increased by about 0.4 percentage point from a base of about 5.3%. The authors do not find clear evidence that overall repayment increased.

- **Economic message.** Regulation can reduce hidden or partially nonsalient fees when markets are imperfectly competitive and consumers do not respond fully to all price components. The CARD Act appears to have reduced consumer costs without a large offset elsewhere.
- **Takeaway.** The paper is an important empirical example of consumer financial protection working as intended: fee regulation lowered borrowing costs, especially for risky and lower-FICO borrowers, while the disclosure nudge had only modest effects.

2 Data and measurement

2.1 Credit card mechanics

A credit card account can be used both for transactions and for borrowing. If a borrower carries a balance, the paper describes interest charges as

$$\text{Interest Charges} = \text{ADB} \times \frac{\text{APR}}{365} \times \text{Days in Billing Cycle}.$$

Here ADB, average daily balance, is the arithmetic mean of end-of-day balances during the billing cycle. This is the denominator used for many of the paper's normalized price and cost measures.

The interpretation is straightforward:

- APR is the annual interest rate;
- APR/365 is the daily interest rate;
- multiplying by ADB and days in the cycle gives the dollar interest charge for that billing cycle.

Consumers who pay off their purchase balance during the grace period generally do not pay interest. Consumers who carry balances pay interest and are more directly affected by borrowing costs and credit card fees.

2.2 The CARD Act timeline

The CARD Act was signed in May 2009 and implemented in phases:

- **Phase 1: August 2009.** Issuers had to provide advance notice for rate increases and major changes in account terms.
- **Phase 2: February 2010.** The major provisions took effect. Over-limit fees required explicit opt-in. Monthly statements had to disclose the cost of making only the minimum payment and the payment amount needed to pay off the balance in 36 months.
- **Phase 3: August 2010.** Penalty fees had to be reasonable and proportional. Late fees were generally capped at \$25 unless one of the previous six payments was also late, in which case a \$35 fee was allowed. Over-limit fees were limited as well.

The empirical analysis focuses mainly on Phase 2 and Phase 3 because these phases directly affected fees and repayment disclosures.

2.3 Data source

The main data set is the **Credit Card Metrics** data assembled by the Office of the Comptroller of the Currency. It covers the near universe of accounts at the largest U.S. credit card issuers.

Key features:

- account-level monthly data;
- approximately 160 million consumer and small business credit card accounts;
- information from the largest banks that issue credit cards;
- monthly data on utilization, contract terms, fees, interest charges, payments, delinquencies, charge-offs, and credit limits;
- the main regression sample runs from April 2008 through December 2011.

The authors drop one small bank that enters and exits the reporting sample and focus on general-purpose consumer cards and small business cards.

2.4 Treatment and control groups

The key comparison is:

Treatment group = consumer credit card accounts,

Control group = small business credit card accounts.

Consumer cards were directly subject to the CARD Act. Small business cards were not. This creates a policy discontinuity across products that are similar in underwriting and risk assessment but different in legal coverage.

2.5 Risk groups

The paper often separates accounts by FICO score at account origination. The key split is:

$$\text{FICO} < 660 \quad \text{versus} \quad \text{FICO} \geq 660.$$

This split matters because low-FICO borrowers paid much higher fees before the law. Therefore, if the CARD Act mattered, the effect should be especially visible among low-FICO accounts.

The paper also uses more detailed FICO bins:

$$< 620, \quad 620 - 659, \quad 660 - 719, \quad 720 - 759, \quad 760 - 799, \quad \geq 800.$$

2.6 Main outcome variables

The main outcomes are:

- **Fees:** over-limit fees, late fees, other fees, and total fees;

- **Interest charges:** actual interest paid, measured as an annualized percent of ADB;
- **Total income:** fees plus interest charges plus interchange income;
- **Costs:** net charge-offs, cost of funds, rewards and fraud expenses, and operational costs;
- **Credit volume:** credit limits, new accounts, and average daily balances;
- **Repayment behavior:** whether the borrower pays the minimum, pays in full, or pays near the 36-month payoff amount.

2.7 Annualized percentage of ADB

Many outcomes are expressed as an annualized percentage of average daily balances. For example, the paper defines fee income as

$$\text{Annualized fee rate} = \left(1 + \frac{\text{Total fees in month}}{\text{ADB}}\right)^{12} - 1.$$

This converts monthly dollar fees into an interest-rate-like measure. If total fees are high relative to the balance, the annualized fee rate will be large.

2.8 Profit components

The paper defines realized profits as total income minus realized costs:

$$\begin{aligned} \text{Realized Profit} = & \text{Interest Charges} + \text{Fees} + \text{Interchange Income} \\ & - \text{Net Charge-offs} - \text{Cost of Funds} \\ & - \text{Rewards/Fraud Expenses} - \text{Operational Costs}. \end{aligned}$$

This profit measure is useful descriptively, but it is not a full ex ante risk-adjusted profitability measure. It uses realized charge-offs during the sample period.

2.9 Payoff variable

For the disclosure nudge, the paper defines T as the number of months needed to pay off the cycle-ending balance if the borrower makes a constant payment and makes no new purchases:

$$T = 1 - \frac{\ln\left(1 - \frac{\text{APR}}{12} \frac{\text{Cycle-Ending Balance} - \text{Payment}}{\text{Payment}}\right)}{\ln\left(1 + \frac{\text{APR}}{12}\right)}.$$

The CARD Act disclosure highlights the payment amount that corresponds to $T = 36$. At the average pre-CARD APR of 16.5%, this payment is about 3.7% of the cycle-ending balance.

3 Empirical specification and interpretation

3.1 Baseline difference-in-differences equation

The baseline model is

$$y_{it} = \alpha_t + \alpha_C \mathbf{1}\{i \in \text{Consumer}\} \quad (1)$$

$$+ \beta_2 \mathbf{1}\{i \in \text{Consumer}\} \mathbf{1}\{t \in \text{Phase2}\} + \beta_3 \mathbf{1}\{i \in \text{Consumer}\} \mathbf{1}\{t \in \text{Phase3}\} \quad (2)$$

$$+ X'_{it} \gamma + \varepsilon_{it}. \quad (3)$$

Here y_{it} is an account-level outcome such as total fees, interest charges, credit limits, or average daily balances.

3.2 Meaning of each term

- α_t : month fixed effects. These absorb aggregate shocks common to consumer and small business cards, such as the financial crisis, monetary policy changes, unemployment changes, and broad deleveraging.
- $\mathbf{1}\{i \in \text{Consumer}\}$: a treatment-group indicator equal to one for consumer credit cards.
- β_2 : the differential change for consumer cards during Phase 2, relative to small business cards and relative to the pre-implementation period.
- β_3 : the differential change for consumer cards during Phase 3, relative to small business cards and relative to the pre-implementation period.
- X_{it} : additional covariates or fixed effects, such as bank-by-FICO group fixed effects in some specifications.

The coefficients β_2 and β_3 are the main causal objects. For example, if the outcome is total fees as a percent of ADB, then $\beta_3 = -5.3$ means that total fees on treated consumer cards fell by 5.3 percentage points of ADB relative to the control group after Phase 3.

3.3 Collapsed estimation

The underlying panel is huge: roughly 160 million accounts over 45 months. Instead of estimating directly on billions of account-month observations, the authors collapse the data to means at the level of

$$\text{bank} \times \text{product type} \times \text{FICO group} \times \text{month}.$$

Product type is based on consumer/small-business status and card categories such as co-branded, oil and gas, affinity, student, or other. The collapsed regressions are weighted so that coefficients can be interpreted as aggregate effects.

3.4 Event-study specification

The paper also uses a flexible month-by-month specification:

$$y_{it} = \alpha_t + \alpha_C \mathbf{1}\{i \in \text{Consumer}\} \quad (4)$$

$$+ \sum_{\tau \neq \text{May 2009}} \delta_\tau \mathbf{1}\{i \in \text{Consumer}\} \mathbf{1}\{t = \tau\} \quad (5)$$

$$+ X'_{it} \gamma + \varepsilon_{it}. \quad (6)$$

This specification is used mainly to show timing:

- whether consumer and small business cards had parallel pre-trends;
- whether outcomes changed sharply around February 2010 or August 2010;
- whether there were anticipatory responses after the law was signed in May 2009.

3.5 Inference

The paper uses two inference approaches.

First, it clusters standard errors by bank-by-product type. This allows errors to be correlated over time within related card products and across accounts whose terms may be jointly set by the bank.

Second, the paper uses permutation tests. The idea is to compare the actual CARD Act effect to placebo effects from randomly assigned treatment months. If the real estimate is extreme relative to placebo estimates, the effect is unlikely to be driven by normal month-to-month variation.

4 Identification and credibility

4.1 Identification logic

The design compares consumer and small business cards before and after the CARD Act:

$$\text{DID} = (\Delta \text{Consumer Cards}) - (\Delta \text{Small Business Cards}).$$

The control group is used to construct the counterfactual path for consumer cards. The identifying assumption is not that consumer and small business cards have the same level of fees or balances. The assumption is that, absent the CARD Act, their changes over time would have been similar.

4.2 Evidence for parallel trends

The paper argues for the parallel-trends assumption in two ways:

- **Graphical evidence.** In the pre-CARD period, consumer and small business cards move similarly in over-limit fees, late fees, interest charges, credit limits, new accounts, and average daily balances.

- **Institutional evidence.** Consumer and small business cards are similar products. Both are often underwritten using personal FICO scores and personal guarantees, and both are issued by the same credit card business units at banks.

4.3 Why this design is useful during the financial crisis

The financial crisis affected default risk, credit limits, consumer borrowing, and bank balance sheets. Month fixed effects absorb shocks that hit consumer and small business cards similarly. The remaining comparison asks whether consumer cards moved differently exactly when the CARD Act provisions became binding.

4.4 Threat: small business cards are not perfect controls

Small business cardholders may have different spending patterns, balances, and business-cycle exposures. This is the biggest conceptual concern with the design.

The paper addresses this concern by:

- showing similar pre-trends;
- conditioning on FICO groups;
- using bank and product-type structure;
- focusing on sharp policy timing;
- interpreting small business cards as a close, not perfect, control group.

4.5 Threat: anticipatory repricing

Banks might have raised interest rates after the law was passed but before the main provisions took effect. The authors explicitly estimate anticipation-period effects and examine interest charges for both existing and new accounts. They find no large anticipatory interest-rate offset.

4.6 Threat: offsetting hidden responses

Issuers could respond to fee limits by increasing other fees, raising interest charges, increasing interchange revenue, reducing rewards, or restricting credit. The paper directly checks these margins.

The evidence shows:

- no meaningful increase in other fees;
- no meaningful increase in interest charges;
- no evidence that total income is preserved through interchange revenue;
- no meaningful restriction of credit limits, new accounts, or ADB.

4.7 What the design does not establish

The paper is strong on short- and medium-run effects. However, the authors emphasize that the data do not fully answer long-run questions, such as:

- whether the regulation changes industry entry or exit;
- whether issuers adjust slowly through multiyear contracts;
- whether banks reduce long-run investments in marketing, IT, scoring models, or product design;
- whether new fee categories appear later.

5 Tables, figures, and results

5.1 Table I: Account-level summary statistics

Read. The average account has an ADB of about \$1,347 and annualized purchase volume of about \$2,138. The average account pays about \$168 in interest charges and about \$58 in total fees per year. Late fees and over-limit fees are major fee categories.

Small business accounts have higher balances and purchase volume than consumer accounts. For example, average ADB is about \$1,923 for small business cards versus about \$1,320 for consumer cards.

Economic message. Credit cards generate meaningful revenue through both explicit interest charges and fees. The fee component is not a small incidental charge; it is economically important, especially for borrowers who revolve balances or miss payments.

TABLE I
SUMMARY STATISTICS USING ACCOUNT-LEVEL DATA (ANNUALIZED \$ PER ACCOUNT)

	Combined sample		Consumer credit cards		Small business credit cards	
	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.
Average daily balance	1,346.55	2,887.52	1,319.55	2,840.56	1,922.76	3,705.39
Purchase volume	2,138.07	10,875.03	1,946.90	9,404.02	6,217.81	25,320.99
Interest charges	168.05	448.29	165.31	438.11	226.51	624.31
Total fees	57.64	230.77	57.72	229.56	55.86	255.35
Annual fee	6.47	63.95	6.58	64.32	4.20	55.57
Balance transfer	3.96	124.93	3.98	125.67	3.58	107.94
Cash advance	4.19	85.67	4.32	86.71	1.43	59.33
Debit suspension	8.10	61.94	8.37	62.82	2.15	37.97
Late fee	24.39	194.06	24.18	191.26	28.83	151.92
Not sufficient funds	0.54	18.20	0.53	17.30	0.70	31.75
Other fees	1.48	52.12	1.46	50.13	1.84	83.92
Over limit	8.51	61.68	8.29	60.01	13.13	90.13
Total charge-offs	174.82	4,619.83	167.65	4,555.38	327.88	6,150.98
Principal	144.14	4,243.99	138.06	4,099.31	273.84	6,614.69
Interest and fees	30.68	710.90	29.59	669.34	54.04	1,314.97
Recovery	6.49	489.39	6.34	453.04	9.71	736.41
Repayment ^a						
Minimum or less (%)	26.1		26.2		23.9	
Full or more (%)	30.2		29.4		43.9	

^aNotes. Values are calculated using all account-months in the sample period. See Online Appendix A for more details. Except for ADB and repayment statistics, values are annualized. The sample period is April 2008–December 2011. All variables are inflation-adjusted to 2012 using the CPI-U.

^bThese statistics are calculated on the sample of account-months with positive cycle-ending balances.

TABLE II
SUMMARY STATISTICS USING PORTFOLIO-LEVEL DATA (ANNUALIZED \$ PER ACCOUNT)

	Mean	Note on construction
Interchange income	42.76	2% of purchase volume
Rewards + fraud expense	29.93	1.4% of purchase volume
Cost of funds	22.49	Share of ADB (time varying)
Operational costs	49.70	Share of ADB (time varying)
Collection	5.81	Share of ADB (time varying)
Marketing + acquisition	7.31	Share of ADB (time varying)
Other operational cost	36.58	Share of ADB (time varying)

Notes. Variables are constructed by combining account-level measures of ADB and purchase volume with information from the portfolio-level data. Values are calculated using all account-months in the sample period. Operational expenses include expenses for marketing and acquisition, collections, servicing, card-holder billing, processing payments, card issuing and administration. See note on construction and Online Appendix A for more details. The sample period is April 2008–December 2011. All variables are inflation-adjusted to 2012 using the CPI-U.

5.2 Table II: Portfolio-level income and costs

Read. The paper constructs account-level measures for items that are available at the portfolio level. Interchange income is approximated as 2% of purchase volume, rewards and fraud expenses as about 1.4% of purchase volume, and funding and operating costs as shares of ADB.

Economic message. The paper is not just estimating revenue effects. It also builds a broader view of credit card profitability by combining interest, fees, interchange, charge-offs, cost of funds, rewards, fraud, and operational costs.

5.3 Table III: Pre-CARD utilization and realized profits by FICO score

Read. Low-FICO borrowers are very different from high-FICO borrowers. Borrowers with FICO below 620 pay about 20.6% of ADB in interest charges and 23.3% of ADB in total fees. Higher-FICO borrowers pay much lower fees, but they generate more purchase volume and interchange income per dollar of balance.

Realized profits are U-shaped in FICO score. The lowest-FICO accounts are profitable despite high charge-offs because they pay very high interest and fee revenue. Accounts in the middle of the FICO distribution are less profitable.

Economic message. The CARD Act targets a market where low-FICO borrowers were paying large penalty fees. The distributional effect is central: the regulation disproportionately benefits the borrowers who were paying the highest fee rates.

TABLE III
PRE-CARD UTILIZATION AND REALIZED PROFITS BY FICO SCORE

	Total	<620	620-659	660-719	720-759	760-799	800+
Percent of accounts	100.0	17.3	12.6	24.6	18.6	19.2	7.6
Panel A: capacity and utilization (annualized \$ per account)							
Credit limit	8,042	2,025	3,546	7,781	11,156	12,400	11,390
ADB	1,410	804	1,469	2,029	1,797	1,110	486
Purchase volume	1,820	730	1,019	1,651	2,306	2,892	2,282
Panel B: realized profits (% of ADB)							
Total income	25.0	45.7	31.5	21.0	16.9	17.1	19.9
Interest charges	14.3	20.6	19.2	15.2	11.8	9.3	7.6
Total fees	7.6	23.3	10.9	4.1	2.5	2.4	2.9
Interchange income	3.2	1.8	1.5	1.7	2.6	5.4	9.5
Total costs	23.4	37.8	30.2	22.5	17.2	15.6	16.8
Net charge-offs	15.6	30.8	23.4	15.8	9.7	6.3	4.7
Cost of funds	2.3	2.4	2.3	2.2	2.2	2.2	2.2
Rewards and fraud	2.2	1.3	1.0	1.2	1.8	3.7	6.5
Operational costs	3.4	3.4	3.4	3.4	3.4	3.4	3.4
Collection	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Marketing & acquisition	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Other operational cost	2.5	2.5	2.5	2.5	2.5	2.5	2.5
Realized profit	1.6	7.9	1.3	-1.6	-0.2	1.5	3.1

Notes: Table shows income and cost components for consumer accounts during the pre-CARD Act period, defined as April 2009-January 2010. The first column shows averages over all accounts; the other columns show averages by FICO score at account origination. Net charge-offs are the sum of principal and interest/fee charge-offs minus recoveries. Panel B shows values as annualized percent of average daily balances. All dollar variables are inflation-adjusted to 2012 using the CPI-U.

If you make no additional charges using this card and each month you pay ...	You will pay off the balance shown in this statement in about ...	And you will end up paying an estimated total of ...
Only the minimum payment	10 years	\$3,284
\$62	3 years	\$2,232 (Savings of \$1,052)

FIGURE I
Payoff Disclosure

Figure provides an example of the disclosure statement on monthly credit card reports required by the CARD Act.

5.4 Figure I: Payoff disclosure

Read. The figure shows the mandated disclosure on monthly credit card statements. It reports the cost of paying only the minimum and compares it with the payment needed to pay off the balance in 36 months.

Economic message. This is the behavioral part of the policy. The regulation tries not only to restrict prices but also to change consumer decisions by making repayment consequences salient.

5.5 Figure II: Pre-CARD profit components by FICO score

Read. Interest charges decline moderately as FICO rises. Fees decline much more sharply. Net charge-offs are highest for low-FICO accounts. Interchange income net of rewards becomes more important for high-FICO accounts.

Economic message. The credit card business model varies across borrower types. Low-FICO accounts produce revenue mainly through borrowing costs and penalty fees. High-FICO accounts produce more revenue through transaction volume and interchange.

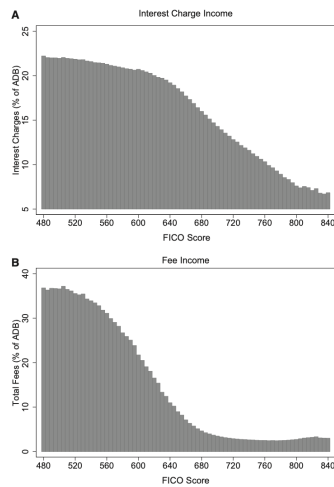


FIGURE II
Pre-CARD Act Profit Components by FICO Score

Figure shows revenue and cost components, realized profits, and number of accounts for consumer credit cards by FICO score at origination binned in groups of five. Revenue and cost components and realized profits are measured as an annualized percentage of ADB. Number of accounts are per reporting month. Realized profits are the difference between revenues (interest charges, fees, and interchange income) and costs (net charge-offs, cost of funds, operational expenses, and fraud and rewards expenses). The sample is restricted to the pre-CARD Act period, defined as April 2008–January 2010.

(continued)

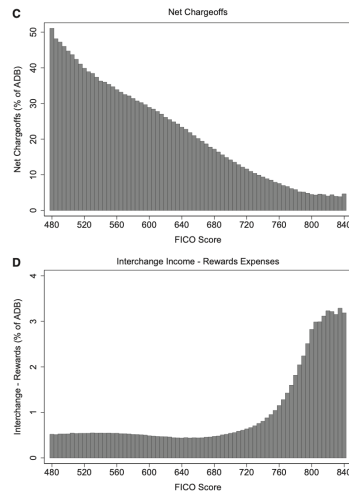


FIGURE II
(continued)

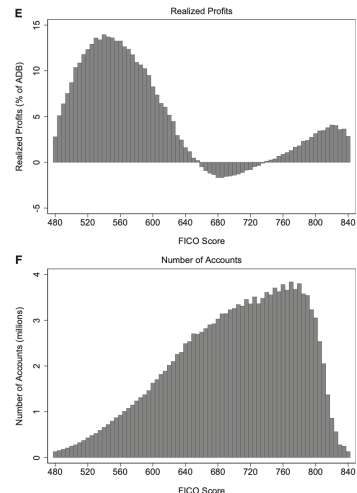


FIGURE II
(continued)

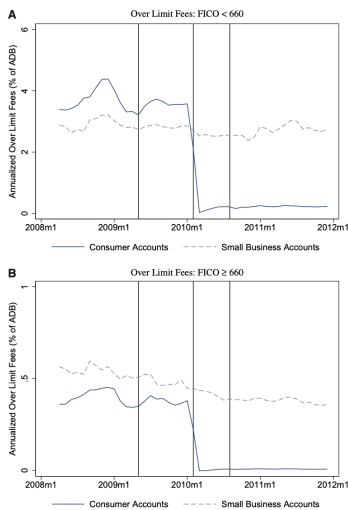


FIGURE III
Over-Limit, Late, and Total Fees

Figures show over-limit fees, late fees, and total fees as an annualized percentage of ADB for account holders with a FICO score less than 660 at origination and a FICO score of at least 660 at origination. The sample period is April 2008–December 2011. Vertical lines are plotted in May 2009, February 2010, and August 2010, the date when the bill was signed and the two key implementation dates of the CARD Act, respectively.

(continued)

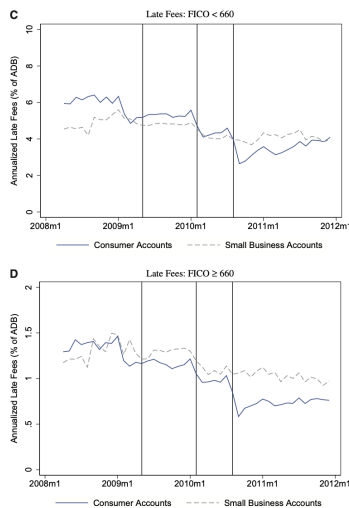


FIGURE III
(continued)

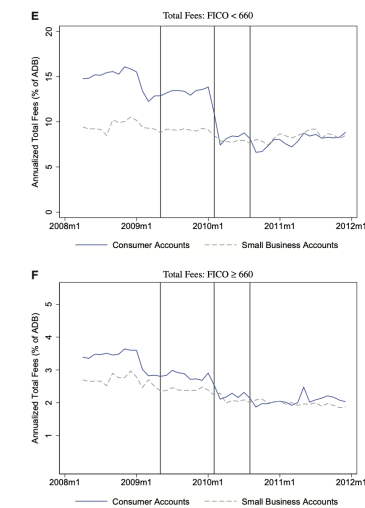


FIGURE III
(continued)

5.6 Figure III: Fee trends around the CARD Act

Read. Over-limit fees for consumer accounts drop sharply to almost zero after the February 2010 implementation of the opt-in rule. Small business cards do not show the same sharp drop. Late fees on consumer cards fall after February 2010 and more sharply after the August 2010 fee caps.

The figure separates low- and high-FICO accounts. The treatment effect is much larger for low-FICO accounts because these accounts paid more fees before the law.

Economic message. This is the main visual evidence that the fee provisions were binding and effective. The timing lines up closely with the policy dates, and the control group does not experience the same discontinuity.

5.7 Table IV: Fee restrictions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Dependent variable: % of ADB							
	Over-limit fees	Late fees	Other fees	Total fees				
Panel A: FICO < 660								
Consumer × phase 2	-3.30** (1.50)	-3.37** (1.52)	-0.61 (0.42)	-0.57 (0.38)	-0.70 (0.46)	-0.62 (0.41)	-4.61** (2.25)	-4.46** (2.22)
	[.03]	[.04]	[.15]	[.14]	[.13]	[.14]	[.05]	[.05]
Consumer × phase 3	-3.37** (1.52)	-3.33** (1.53)	-1.41* (0.76)	-1.47** (0.59)	-0.57 (0.45)	-0.52 (0.37)	-5.35** (2.67)	-5.32** (2.55)
	[.03]	[.03]	[.07]	[.04]	[.21]	[.17]	[.05]	[.04]
Pre-CARD Act, consumer mean	3.77	3.77	5.85	5.85	5.05	5.05	14.68	14.68
R-squared	0.32	0.73	0.19	0.84	0.04	0.87	0.11	0.84
Number of observations	3,447	3,447	3,447	3,447	3,447	3,447	3,447	3,447
Panel B: FICO > 660								
Consumer × phase 2	-0.30*** (0.11)	-0.30*** (0.11)	-0.07 (0.07)	-0.08 (0.07)	-0.08 (0.10)	-0.07 (0.10)	-0.45*** (0.14)	-0.45*** (0.15)
	[.01]	[.01]	[.32]	[.26]	[.43]	[.49]	[.00]	[.00]
Consumer × phase 3	-0.25** (0.12)	-0.26** (0.12)	-0.25*** (0.08)	-0.30*** (0.10)	0.06 (0.08)	0.07 (0.07)	-0.45** (0.20)	-0.49** (0.20)
	[.04]	[.04]	[.00]	[.00]	[.46]	[.32]	[.03]	[.02]
Pre-CARD Act, consumer mean	0.40	0.40	1.27	1.27	1.50	1.50	3.17	3.17
R-squared	0.23	0.69	0.11	0.73	0.19	0.52	0.16	0.72
Number of observations	6,986	6,986	6,986	6,986	6,986	6,986	6,986	6,986

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Dependent variable: % of ADB							
	Over-limit fees	Late fees	Other fees	Total fees				
Controls: all panels								
Main effects:								
Consumer card FE	X	X	X	X	X	X	X	X
Month FE	X	X	X	X	X	X	X	X
Additional covariates								
Bank FE × FICO score group FE	X		X		X		X	

Note: Table shows coefficients from difference-in-differences regressions that compare fees for consumer credit cards (treatment group) and small business cards (control group) during the different phases of the CARD Act implementation. The dependent variables are shown as an annualized percent of ADB and are calculated in the same manner as the values in Panel B of Table II. We define phase 2 as March 2010–August 2010 and phase 3 as the months after August 2010. The period prior to February 2010 is the omitted group, so the coefficients can be interpreted as the differential effect relative to the preimplementation period. The sample period is April 2008–December 2011. The regressions are estimated on data aggregated to the bank × product type × FICO score group × month level, and weighted by total ADB in each group. A product type is defined as the intersection of the consumer card indicator and whether the card is co-branded, oil and gas, affinity, student, or other. FICO score groups are <625, 625–659, 660–719, 720–759, 760–799, and ≥ 800. Standard errors clustered by bank × product type are shown in parentheses and the associated p-values are shown in brackets. There are 66 such clusters in Panel A and 47 clusters in Panel B. Significance levels: * $p < .10$, ** $p < .05$, *** $p < .01$.

Read. For accounts with FICO below 660, the preferred Phase 3 estimates show:

- over-limit fees fall by about 3.3 percentage points of ADB;
- late fees fall by about 1.5 percentage points of ADB;
- other fees do not increase enough to offset the decline;
- total fees fall by about 5.3 percentage points of ADB.

For accounts with FICO at least 660, total fees fall by about 0.5 percentage point of ADB.

Economic message. The fee provisions worked almost exactly on the margins they targeted. The biggest effects are over-limit fees and late fees. There is no evidence that issuers simply relabeled the lost revenue into other fees in the short run.

5.8 Figure IV: Utilization and late-payment mechanisms

Read. Panel A shows that the share of accounts over the credit limit declines modestly, but not enough to explain the collapse in over-limit fee revenue. Panel B shows that many accounts still exceed limits after the law. Panels C and D show that the incidence of late payments does not rise; instead, the dollar amount of late fees changes, especially from \$40 fees to \$25 and \$35 fees.

Economic message. The regulation reduced the price charged for certain events rather than simply stopping the events. Borrowers were still allowed to go over limits or pay late, but issuers could not charge the same penalty fees.

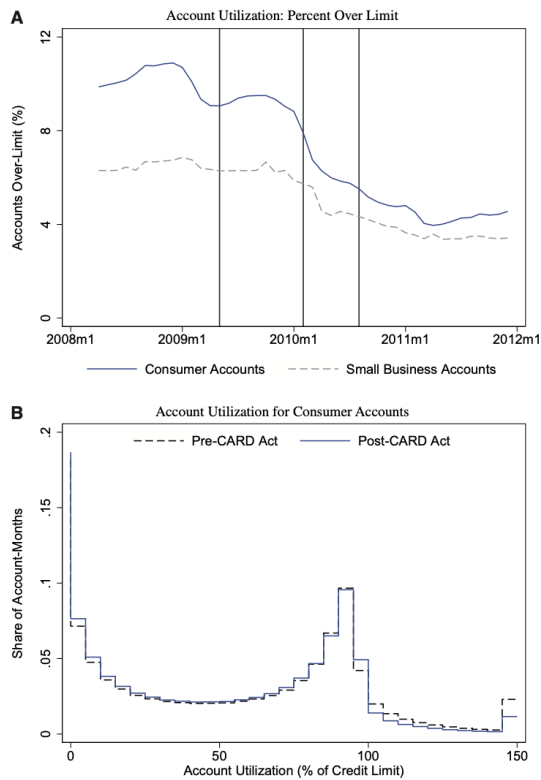


FIGURE IV

Account Utilization and Late Payments

Figures show information on account utilization and late payments. Panel A shows the percentage of consumer and small business accounts with cycle-ending balances exceeding the credit limit in each month. Panel B shows the distribution of cycle-ending balances as a share of the credit limit for consumer credit cards in the year before (dashed line) and after (solid line) the February 2010 CARD Act implementation date.

(continued)

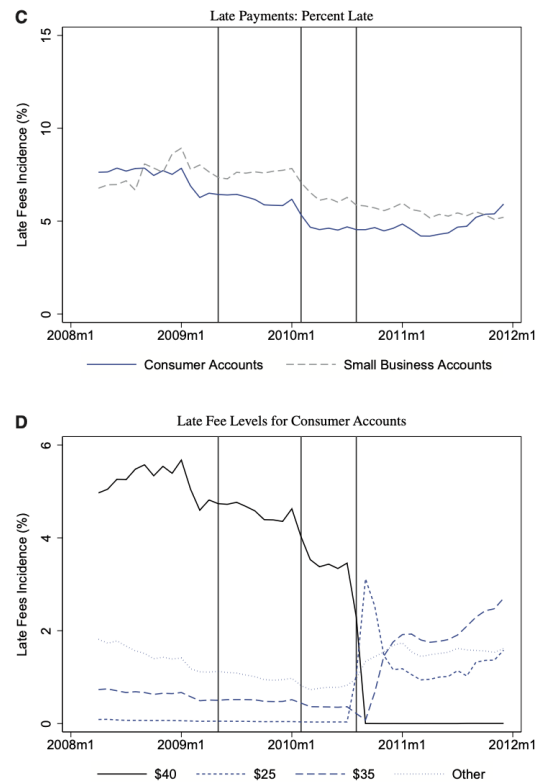


FIGURE IV

(continued)

Panel C shows the percentage of consumer and small business accounts with positive late fees in each month. Panel D shows the percent of consumer accounts with late fees of \$25, \$35, \$45, and an "other" category that combines all other positive levels. The \$25 category is defined as accounts with late fees in the \$21–\$25 range, the \$35 category as accounts with late fees in the \$31–\$35 range, and the \$40 category as accounts with late fees in the \$36–\$40 range. The sample period is April 2008–December 2011. Vertical lines are plotted in May 2009, February 2010, and August 2010, the date when the bill was signed and the two key implementation dates of the CARD Act, respectively.

5.9 Figure V: Pre-CARD payoff behavior

Read. The probability of paying the full balance rises strongly with FICO score. The probability of paying the minimum or less falls with FICO score. Low-FICO borrowers are much more likely to make low payments and carry debt.

Economic message. The repayment nudge is aimed at borrowers who may be under-repaying or inattentive to the long-run cost of minimum payments. The figure documents large heterogeneity in repayment behavior.

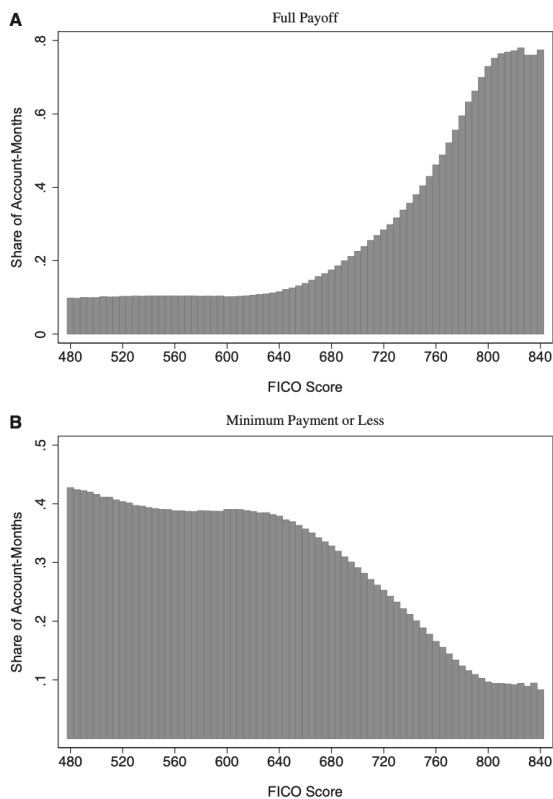


FIGURE V

Pre-CARD Act Payoff Distribution

Figure shows payoff behavior by FICO score at origination in the pre-CARD Act period, defined as April 2008–January 2010. Panel A shows the share of account-months making the full payment. Panel B shows the share of account-months making the minimum payment or less. The sample excludes accounts that have a zero cycle-ending balance.

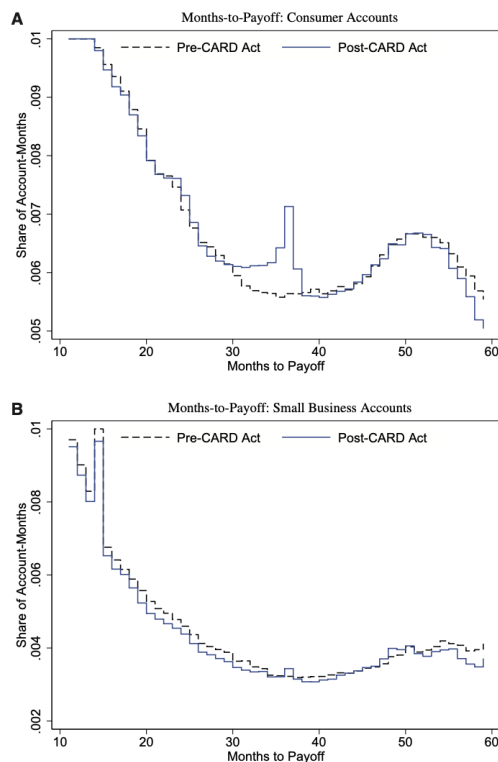


FIGURE VI

Months-to-Payoff (T)

Figure plots distributions of months-to-payoff (T) in the year before (dashed line) and after (solid line) the February 2010 CARD Act implementation date. Months-to-payoff (T) is the number of months it would take to pay off the cycle-ending balance if the account holder makes constant payments and makes no new purchases, and is calculated using equation (3). We present the distribution for T between 10 and 60. The share of account-months is top-coded at 1% to focus on the distribution around the CARD Act target payoff amount ($T = 36$). Panel A shows the distribution for consumer credit cards, Panel B for small business credit cards.

5.10 Figure VI: Months-to-payoff distribution

Read. The figure plots the distribution of T , the implied months to pay off the balance. After the CARD Act, consumer accounts show a small increase in mass near $T = 36$. Small business accounts do not show the same change.

Economic message. The disclosure nudge changed behavior for a small group of borrowers. It created bunching near the disclosed 36-month payoff amount, but the response is modest.

TABLE V
PAYOFF NUDGE: DIFFERENCE-IN-DIFFERENCES REGRESSIONS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Dependent variable: percent of payments								
	Nudge range ($30 \leq T \leq 38$)			Less than nudge range ($T > 38$)			Minimum payment		
Consumer $\times$ phase 2	0.39** (0.19) [.05]	0.38** (0.17) [.03]	0.75** (0.34) [.03]	0.03 (1.23) [.98]	-0.15 (1.07) [.89]	0.46 (1.29) [.72]	-0.22 (0.36) [.54]	-0.25 (0.40) [.54]	-0.35 (0.47) [.46]
Controls									
Main effects									
Consumer card FE	X	X	X	X	X	X	X	X	X
Month FE	X	X	X	X	X	X	X	X	X
Additional covariates									
Bank FE $\times$ FICO score group FE		X	X		X	X		X	X
Sample restrictions									
Excluding accounts with balance < \$1,000			X			X			X
Pre-CARD Act, consumer mean	5.31	5.31	7.19	35.35	35.35	47.78	12.95	12.95	14.97
R-squared	0.08	0.87	0.78	0.06	0.90	0.88	0.06	0.87	0.88
Number of observations	10,433	10,433	10,433	10,433	10,433	10,433	10,433	10,433	10,433

Note: Table shows estimates from difference-in-differences regressions of payoff behavior for consumer credit cards (treatment group) and small business cards (control group) around the implementation of the payoff nudge. In columns (1)-(3), the dependent variable is the percent of account holders that make a payment corresponding to the target CARD Act payment nudge amount ($30 \leq T \leq 38$). In columns (4)-(6), the dependent variable is the percent of account holders that make payments smaller than the CARD Act payment nudge amount ($T > 38$). In columns (7)-(9), the dependent variable is the percent of account holders that make exactly the minimum payment. The sample period is April 2008–December 2011. The regressions are estimated on data aggregated to the bank $\times$ product type $\times$ FICO score group $\times$ month level, and weighted by the number of accounts in each group. A product type is defined as the intersection of the consumer card indicator and whether the card is co-branded, oil and gas, affinity, student, or other. FICO score groups are <620, 620–659, 660–719, 720–759, 760–799, and ≥ 800 . In columns (3), (6), and (9), the sample is restricted to accounts with a cycle-ending balance larger than \$1,000. Standard errors clustered by bank $\times$ product type are shown in parentheses and the associated p -values are shown in brackets. There are 47 such clusters in the sample. Significance levels: * $p < .10$, ** $p < .05$, *** $p < .01$.

5.11 Table V: Payoff nudge

Read. The nudge increases the share of accounts making payments close to the 36-month amount by about 0.4 percentage point from a pre-CARD mean of 5.3%. The effect is larger, about 0.75 percentage point, when restricting to accounts with balances above \$1,000.

The paper does not find precise evidence that the nudge shifted borrowers from lower payments to the 36-month payment. It also does not find evidence that the disclosure caused more borrowers to anchor on the minimum payment.

Economic message. Disclosure can move behavior at the exact disclosed target, but the aggregate effect is small. The authors estimate an upper bound of about \$57 million per year in savings, far smaller than the estimated fee savings.

5.12 Figure VII: Interest charges and credit volume

Read. For low-FICO accounts, consumer and small business cards have similar pre-trends in interest charges, credit limits, new accounts, and ADB. After the CARD Act, there is no clear differential increase in interest charges and no clear differential contraction in credit volume for consumer cards.

Economic message. The figure addresses the central criticism of fee regulation: that banks will raise other prices or reduce credit access. The visual evidence does not support that criticism in this setting.

5.13 Table VI: Interest-charge offsets and total income

Read. For low-FICO accounts, the preferred Phase 3 estimate for interest charges is close to zero. The implied offset is about -0.09 , meaning there is no evidence of a positive offset. The paper can reject an offset above 61% at the 95% confidence level.

Total income falls by about 6.5 percentage points of ADB, close to the fee decline. Costs excluding charge-offs do not fall enough to offset the loss of income.

Economic message. The fee savings appear to be real savings for consumers, not just a relabeling of charges. Banks' revenue falls, which is exactly what would happen if the regulation reduced nonsalient fee extraction.

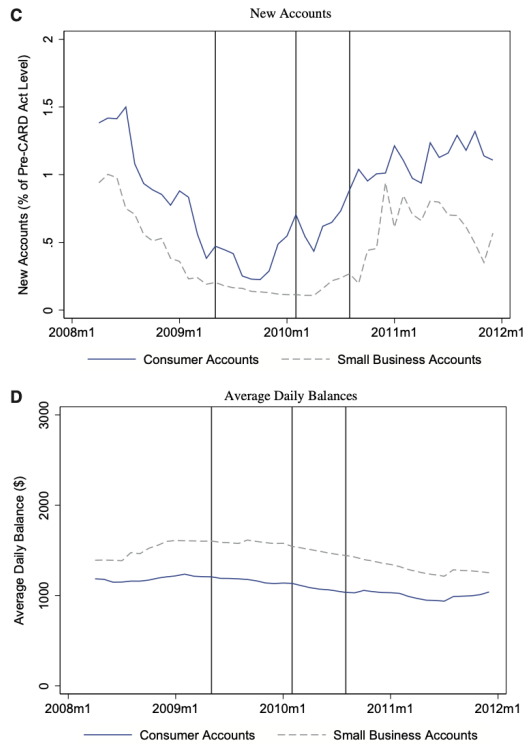


FIGURE VII
(continued)

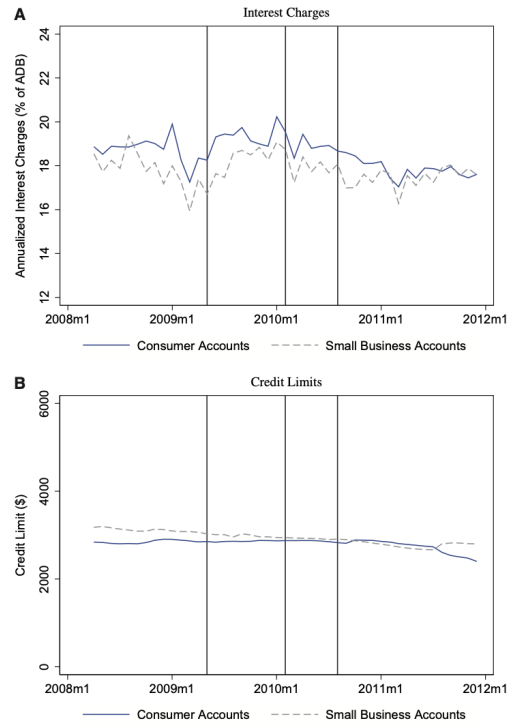


FIGURE VII

Interest Charges and Credit Volume: FICO < 660

Panel A shows interest charges as an annualized percentage of ADB. Panel B shows credit limits. Panel C shows new accounts measured as a percentage of the average total pre-CARD Act number of accounts. Panel D shows average daily balances. All panels focus on account holders with a FICO score below 660 at origination, and display monthly averages for consumer and small business credit cards. The sample period is April 2008–December 2011. Vertical lines are plotted in May 2009, February 2010, and August 2010, the date when the bill was signed and the two key implementation dates of the CARD Act, respectively.

TABLE VI
INTEREST CHARGE OFFSET: DIFFERENCE-IN-DIFFERENCE REGRESSIONS (FICO < 660)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Interest charges				Total income			
	Offset				Costs excluding charge-offs			
Consumer × anticipation	0.25 (1.40)	0.29 (1.40)	0.05 (0.26)	0.06 (0.26)	-0.37 (1.51)	0.01 (1.47)	-0.06 (0.17)	-0.05 (0.15)
Consumer × phase 2	1.86 (2.31)	1.84 (2.26)	1.86 (0.43)	1.84 (0.42)	1.81 (2.27)	1.09 (2.20)	1.71 (0.25)	1.71 (0.23)
Consumer × phase 3	1.92 (2.03)	1.94 (1.89)	1.92 (0.38)	1.94 (0.36)	1.95 (2.72)	1.05 (2.36)	1.96 (0.29)	1.96 (0.24)
Controls	1.87 (1.87)	1.88 (1.88)	1.87 (1.87)	1.88 (1.88)	1.92 (1.92)	1.01 (1.01)	1.94 (1.94)	1.01 (1.01)
Main effects								
Consumer card FE	X	X	X	X	X	X	X	X
Month FE	X	X	X	X	X	X	X	X
Additional covariates								
Bank FE × FICO score group FE	X	X	X	X	X	X	X	X
Pre-CARD Act, consumer mean	19.14	19.14	N/A	N/A	35.40	35.40	3.65	3.65
R-squared	0.06	0.78	N/A	N/A	0.05	0.50	0.30	0.30
Number of observations	3,447	3,447	3,447	3,447	3,447	3,447	3,447	3,447

Note: Table shows coefficients from difference-in-differences regressions that compare outcomes for consumer credit cards (treatment group) and small business cards (control group) during the different phases of the CARD Act implementation. We define the Anticipation period as the months between the passage of the bill in May 2009 and the implementation of phase 2 in February 2010. We define phase 2 as March 2010 to August 2010 and phase 3 as the months after August 2010. The period prior to April 2009 is the omitted group, so the coefficients can be interpreted as the differential effect relative to pre-CARD Act period. The sample is restricted to accounts with FICO scores below 660 at origination. The sample period is April 2008–December 2011. The regressions are estimated on data aggregated to the bank × product type × FICO score group × month level, and weighted by total ADB in each group. A product type is defined as the intersection of the consumer card indicator and whether the card is co-branded, oil and gas, affinity, student, or other. FICO score groups are <620, 620–659, 660–719, 720–759, 760–799, and ≥800. Standard errors clustered by bank × product type are shown in parentheses and the associated p-values are shown in brackets. There are 46 such clusters in the sample. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE VI
INTEREST CHARGE OFFSET: DIFFERENCE-IN-DIFFERENCE REGRESSIONS (FICO < 660)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Interest charges				Total income			
	Offset				Costs excluding charge-offs			
Consumer × anticipation	0.25 (1.40)	0.29 (1.40)	0.05 (0.26)	0.06 (0.26)	-0.37 (1.51)	0.01 (1.47)	-0.06 (0.17)	-0.05 (0.15)
Consumer × phase 2	1.86 (2.31)	1.84 (2.26)	1.86 (0.43)	1.84 (0.42)	1.81 (2.27)	1.09 (2.20)	1.71 (0.25)	1.71 (0.23)
Consumer × phase 3	1.92 (2.03)	1.94 (1.89)	1.92 (0.38)	1.94 (0.36)	1.95 (2.72)	1.05 (2.36)	1.96 (0.29)	1.96 (0.24)
Controls	1.87 (1.87)	1.88 (1.88)	1.87 (1.87)	1.88 (1.88)	1.92 (1.92)	1.01 (1.01)	1.94 (1.94)	1.01 (1.01)
Main effects								
Consumer card FE	X	X	X	X	X	X	X	X
Month FE	X	X	X	X	X	X	X	X
Additional covariates								
Bank FE × FICO score group FE	X	X	X	X	X	X	X	X
Pre-CARD Act, consumer mean	19.14	19.14	N/A	N/A	35.40	35.40	3.65	3.65
R-squared	0.06	0.78	N/A	N/A	0.05	0.50	0.30	0.30
Number of observations	3,447	3,447	3,447	3,447	3,447	3,447	3,447	3,447

Note: Table shows coefficients from difference-in-differences regressions that compare outcomes for consumer credit cards (treatment group) and small business cards (control group) during the different phases of the CARD Act implementation. We define the Anticipation period as the months between the passage of the bill in May 2009 and the implementation of phase 2 in February 2010. We define phase 2 as March 2010 to August 2010 and phase 3 as the months after August 2010. The period prior to April 2009 is the omitted group, so the coefficients can be interpreted as the differential effect relative to pre-CARD Act period. The sample is restricted to accounts with FICO scores below 660 at origination. The sample period is April 2008–December 2011. The regressions are estimated on data aggregated to the bank × product type × FICO score group × month level, and weighted by total ADB in each group. A product type is defined as the intersection of the consumer card indicator and whether the card is co-branded, oil and gas, affinity, student, or other. FICO score groups are <620, 620–659, 660–719, 720–759, 760–799, and ≥800. Standard errors clustered by bank × product type are shown in parentheses and the associated p-values are shown in brackets. There are 46 such clusters in the sample. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE VII
CREDIT VOLUME: DIFFERENCE-IN-DIFFERENCES REGRESSIONS (FICO < 660)

	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent variable:					
	Credit limits (\$)		New accounts (% of pre-CARD Act level)		Average daily balances (\$)	
Consumer × anticipation	124.31 (153.63) [.42]	40.62 (163.42) [.80]	−0.19 (0.11) [.09]	−0.17 (0.11) [.13]	−105.66* (59.32) [.08]	−128.09** (58.28) [.03]
Consumer × phase 2	229.37 (285.74) [.43]	101.72 (295.80) [.73]	0.02 (0.17) [.90]	0.02 (0.12) [.87]	−98.13 (142.37) [.49]	−128.14 (136.72) [.35]
Consumer × phase 3	291.23 (326.38) [.38]	113.95 (309.64) [.71]	0.08 (0.21) [.71]	0.09 (0.23) [.70]	9.30 (190.25) [.96]	−39.28 (176.21) [.82]
Controls						
Main effects						
Consumer card FE	X	X	X	X	X	X
Month FE	X	X	X	X	X	X
Additional covariates						
Bank FE × FICO score group FE		X		X		X
Pre-CARD Act, consumer mean	2,807.88	2,807.88	0.76	0.76	1,159.29	1,159.29
R-squared	0.00	0.74	0.04	0.24	0.03	0.79
Number of observations	3,447	3,447	3,447	3,447	3,447	3,447

Notes. Table shows coefficients from difference-in-differences regressions that compare measures of credit volume for consumer credit cards (treatment group) and small business cards (control group) during the different phases of the CARD Act implementation. Columns (1) and (2) show regressions with credit limits as the dependent variable. Columns (3) and (4) show regressions with new accounts as a percentage of the pre-CARD Act average number of accounts as the dependent variable. Columns (5) and (6) shows regressions with average daily balances as the dependent variable. We define the anticipation period as the months between the passage of the bill in May 2009 and the implementation of phase 2 in February 2010. We define phase 2 as March 2010–August 2010 and phase 3 as the months after August 2010. The period prior to April 2009 is the omitted group, so the coefficients can be interpreted as the differential effect relative to pre-CARD Act period. The sample is restricted to accounts with FICO scores below 660 at origination. The sample period is April 2008–December 2011. The regressions are estimated on data aggregated to the bank × product type × FICO score group × month level, and weighted by the number of accounts in each group. A product type is defined as the interaction of the consumer card indicator and whether the card is co-branded, oil and gas, affinity, student, or other. FICO score groups are <620, 620–659, 660–719, 720–759, 760–799, and ≥800. Standard errors clustered by bank × product type are shown in parentheses and the associated *p*-values are shown in brackets. There are 46 such clusters in the sample. Significance levels: * *p* < .10, ** *p* < .05, *** *p* < .01.

5.14 Table VII: Credit volume

Read. The outcomes are credit limits, new accounts, and average daily balances for low-FICO accounts. The preferred Phase 3 estimates do not show a reduction in credit limits, originations, or balances. The point estimate for credit limits is actually positive, though not statistically significant.

Economic message. The paper finds no evidence that issuers responded to fee regulation by rationing credit. In the short to medium run, consumers received lower fees without losing access to credit on the measured margins.

6 Short summary

6.1 What the paper asks

The paper asks whether regulating credit card fees and disclosures can reduce consumer borrowing costs without causing banks to undo the regulation through other prices or credit rationing.

6.2 What the paper does

The authors use monthly account-level data from major U.S. credit card issuers and compare consumer cards affected by the CARD Act to small business cards that were not covered by the law. The main design is difference-in-differences.

6.3 What the paper finds

The fee rules significantly reduce over-limit fees and late fees. The total fee decline is about 5.3 percentage points of ADB for low-FICO borrowers and about 0.5 percentage point for higher-FICO borrowers. Aggregated across the market, the estimated consumer savings are about \$11.9 billion per year.

The disclosure nudge produces small bunching near the 36-month payoff amount but little evidence of large aggregate repayment changes.

6.4 Why the result is credible

The timing of the fee decline matches the policy implementation dates. Consumer and small business cards have similar pre-trends. The authors check multiple possible offsets: interest charges, other fees, interchange, costs, credit limits, new accounts, and ADB.

6.5 Why it matters

The paper directly addresses a core debate in household finance and regulation. It shows that consumer protection rules can matter quantitatively, especially when they target fees that are less salient to consumers and when firms have market power.

7 Conclusion

7.1 Core conclusion

The CARD Act reduced credit card fees substantially, especially for lower-FICO borrowers, and did not produce a detectable offset through higher interest charges or reduced credit access during the sample period.

7.2 Economic intuition

If consumers do not fully perceive or react to all fee components, and if card issuers have some market power, then hidden or nonsalient fees can be above the level that would prevail under fully salient pricing. Regulation can reduce those fees without mechanically forcing a one-for-one increase in other prices.

7.3 Policy implication

The paper supports the view that consumer financial protection can generate large benefits when it targets fee structures that exploit inattention, limited salience, or behavioral biases. However, disclosure nudges may have smaller effects than direct price regulation.

7.4 Limits and future research

The paper has about two years of post-CARD data. It cannot fully evaluate long-run changes in entry, exit, product innovation, IT investment, credit-scoring models, or new fee design. It also relies on small business cards as the control group, which is credible but not perfect.

7.5 Takeaway

The main lesson is simple: in this market, fee regulation had large and measurable consumer benefits, while the behavioral disclosure had only modest effects. The CARD Act lowered costs where fees had been highest and did so without clear evidence of offsetting price increases or credit contraction.